

# HydroTrack Project Guide

A simple presentation guide for the Water Fountain Cleanliness Monitoring System

HydroTrack supports Sustainable Development Goal 6 by helping a campus monitor clean drinking water access, maintain refill stations, and encourage reusable bottles instead of single-use plastic bottles.

|                    |                                                                          |
|--------------------|--------------------------------------------------------------------------|
| Project            | HydroTrack                                                               |
| Main idea          | Monitor water fountains and refill stations in NU Manila Main Building   |
| Audience           | Guests, inspectors, technicians, and admins                              |
| Tech stack         | Spring Boot, Thymeleaf, Spring Security, JPA, MySQL                      |
| Presentation focus | Clean water, simple guest access, maintenance tracking, and SDG 6 impact |

# 1. Big Picture

HydroTrack is a web application for checking and managing water fountains in a building. The main goal is to make clean drinking water easier to trust and easier to find. Instead of relying on paper records or verbal reports, the system stores fountain status, water quality readings, filter schedules, and maintenance logs in one place.

## How to explain it in one sentence

HydroTrack helps students and staff find safe refill stations while helping authorized staff monitor quality, filters, and maintenance work.

## Connection to SDG 6

- SDG 6 means Clean Water and Sanitation.
- HydroTrack supports SDG 6 by making drinking water access visible and monitored.
- The app also supports less plastic bottle use because people can confidently refill reusable bottles.
- The system turns small daily checks into records that the school can act on.

# 2. User Roles

The app uses roles to decide what each person can see or do. This keeps the guest experience simple while protecting operational features from accidental changes.

| Role           | What they can do                                          | Presentation explanation                                              |
|----------------|-----------------------------------------------------------|-----------------------------------------------------------------------|
| Guest / Viewer | View simple water overview and find stations.             | For students or visitors who only need to know where to refill.       |
| Inspector      | Record water quality readings.                            | For people checking pH, turbidity, TDS, temperature, and cleanliness. |
| Technician     | Manage fountains, filter schedules, and maintenance logs. | For people who repair, clean, and maintain the stations.              |
| Admin          | Manage users and system data.                             | For the person responsible for accounts and full system control.      |

# 3. Main User Flows

## Before login

- The landing page introduces HydroTrack and the SDG 6 purpose.
- Users can sign in, create a guest account, or reset a password.
- Guest accounts are automatically created as VIEWER accounts.

## After login

- Everyone lands on the control center menu.
- Guests see a lighter interface: Water Overview and Find Stations.
- Staff users see operational pages depending on role.

- The HydroTrack logo in the sidebar returns users to the control center menu.
- The top-right theme icon switches between dark mode and light mode and remembers the choice.

## 4. How the Code Is Organized

HydroTrack follows the common Spring Boot MVC pattern. Think of it as layers: pages send requests to controllers, controllers call services, services use repositories, and repositories talk to the database.

Browser page -> Controller -> Service -> Repository -> MySQL database

| Folder                                     | Purpose                                                                        |
|--------------------------------------------|--------------------------------------------------------------------------------|
| src/main/java/com/hydrotrack/controller    | Receives web requests and chooses which page to show.                          |
| src/main/java/com/hydrotrack/service       | Contains business logic such as creating users or calculating dashboard stats. |
| src/main/java/com/hydrotrack/repository    | JPA database access layer. These interfaces query MySQL.                       |
| src/main/java/com/hydrotrack/model         | Entity classes that match database tables.                                     |
| src/main/java/com/hydrotrack/config        | Security, startup data, and shared helpers.                                    |
| src/main/resources/templates               | Thymeleaf HTML pages shown to users.                                           |
| src/main/resources/static                  | CSS, JavaScript, and images.                                                   |
| src/main/resources/schema.sql and data.sql | Database structure and starter data.                                           |

## 5. Important Backend Files

| File                        | What it does                                                                            |
|-----------------------------|-----------------------------------------------------------------------------------------|
| HydroTrackApplication.java  | Starts the Spring Boot application.                                                     |
| SecurityConfig.java         | Controls login, logout, public pages, and role permissions.                             |
| AuthController.java         | Handles login page, guest registration, and forgot-password flow.                       |
| UserController.java         | Admin user management: create, enable or disable, and delete accounts.                  |
| DashboardController.java    | Loads summary data for the dashboard.                                                   |
| FountainController.java     | Shows station list and station detail pages; staff can create or edit stations.         |
| WaterQualityController.java | Shows and records water quality checks.                                                 |
| ScheduleController.java     | Manages filter replacement schedules.                                                   |
| MaintenanceController.java  | Manages maintenance logs.                                                               |
| GlobalControllerAdvice.java | Adds shared page data, including request path and friendly floor labels like 1st Floor. |

## 6. Important Service Files

Services are where most of the app decisions happen. Controllers stay thinner because they pass work into these classes.

| Service             | Simple explanation                                                                                    |
|---------------------|-------------------------------------------------------------------------------------------------------|
| UserService         | Loads users for login, creates accounts, hashes passwords, resets passwords, and changes user status. |
| FountainService     | Creates, updates, deletes, and filters fountains by floor or status.                                  |
| WaterQualityService | Stores and reads water quality records for each fountain.                                             |
| ScheduleService     | Tracks filter schedules, overdue filters, and upcoming replacements.                                  |
| MaintenanceService  | Stores cleaning, repair, filter-change, and inspection logs.                                          |
| ReportService       | Counts important dashboard numbers such as operational stations and overdue filters.                  |

## 7. Database Tables

Each model class maps to a table in MySQL. The schema file creates the tables, and the data file fills them with starting roles, users, the building, stations, quality checks, schedules, and maintenance logs.

| Table                 | Stores                                                                                    |
|-----------------------|-------------------------------------------------------------------------------------------|
| roles                 | Role names such as ADMIN, TECHNICIAN, INSPECTOR, and VIEWER.                              |
| users                 | Login accounts, emails, password hashes, full names, and active status.                   |
| user_roles            | Which roles belong to each user.                                                          |
| buildings             | Building name, code, address, floors, and coordinates.                                    |
| fountains             | Station code, type, floor number, location, status, install date, and last filter change. |
| water_quality_records | pH, turbidity, TDS, temperature, cleanliness rating, notes, and inspector.                |
| filter_schedules      | Filter type, interval, next replacement date, priority, and status.                       |
| maintenance_logs      | Cleaning, repair, inspection, filter changes, technician, and cost.                       |
| audit_logs            | A place for tracking changes made in the system.                                          |

## 8. Security and Account Features

### Login and permissions

- Spring Security protects all pages except public assets, login, register, forgot-password, and error pages.
- Users log in through the custom login page at /login.
- Successful login redirects to /control-center.
- Passwords are stored as BCrypt hashes, not plain text.

### Create account

- The register form asks for full name, email, username, password, and confirm password.

- The controller checks that both passwords match.
- UserService creates the account with the VIEWER role, which is the guest role.

## Forgot password

- The form asks for username, email, new password, and confirm password.
- The system only resets the password if the username and email match the same account.
- The new password is encoded with BCrypt before it is saved.

## 9. Frontend Pages

| Template              | What users see                                                                   |
|-----------------------|----------------------------------------------------------------------------------|
| login.html            | Public landing page focused on SDG 6, clean water, and reducing plastic bottles. |
| register.html         | Guest account creation page.                                                     |
| forgot-password.html  | Password reset page.                                                             |
| layout.html           | Shared sidebar, logged-in layout, alerts, theme toggle, and navigation.          |
| control-center.html   | Menu after login. Guests see simple options; staff see operational options.      |
| dashboard.html        | Guest overview or staff dashboard depending on role.                             |
| fountains/list.html   | Station finder and fountain list.                                                |
| fountains/detail.html | Station details. Guests see simple guidance; staff see deeper records.           |
| admin/users.html      | Admin page for managing accounts, including delete account.                      |

## 10. Guest Experience

Guests should not need to understand maintenance operations. The guest UI uses simpler words and hides complicated pages.

- Dashboard becomes Water Overview.
- Fountains becomes Find Stations.
- Status labels become Ready to use, Check before using, Unavailable, or Under repair.
- Maintenance, schedules, water quality forms, and admin pages are hidden from guests.
- Station details show a short guide instead of operational history tables.

## 11. Staff and Admin Experience

### Inspector

The inspector can add quality records. These records include pH level, turbidity, TDS, temperature, cleanliness rating, and notes.

## Technician

The technician can manage fountains, schedules, and maintenance logs. This supports repairs, filter changes, cleaning, and inspections.

## Admin

The admin can manage user accounts. The admin can create users, enable or disable users, and delete other accounts. The current signed-in admin account is protected from deleting itself.

## 12. UI Improvements

- The public landing page is mission-focused and explains HydroTrack before login.
- Icons were changed from emojis to cleaner CSS mask icons for a more professional look.
- Dark mode and light mode are saved in localStorage so the chosen theme stays after page changes.
- The theme switch is an icon button in the top-right corner after signing in.
- Clicking HydroTrack in the sidebar returns users to the control center menu.
- Floor labels now show 1st Floor through 8th Floor instead of F1 through F8.

## 13. How to Present the System

### Suggested demo order

- Start at the landing page and explain SDG 6 and plastic bottle reduction.
- Create or show a guest account and log in as a guest.
- Show the simple guest dashboard and station finder.
- Log in as admin or staff to show the operational side.
- Show how water quality records, filters, maintenance, and user management support the same clean-water goal.
- End by explaining how records help the school decide what station needs action.

### Short speaking script

HydroTrack is a water fountain cleanliness monitoring system for NU Manila. Guests can quickly find available refill stations, while staff can monitor water quality, filters, and maintenance work. The goal is to support SDG 6 by improving access to clean water and encouraging reusable bottles instead of single-use plastic bottles.

## 14. Common Questions and Simple Answers

| Question        | Simple answer                                                                              |
|-----------------|--------------------------------------------------------------------------------------------|
| Why use roles?  | Roles keep the system safe. Guests can view, while staff can update records.               |
| Why use BCrypt? | It protects passwords by saving a secure hash instead of the actual password.              |
| Why use MySQL?  | The app needs permanent storage for users, stations, quality records, schedules, and logs. |

| Question                            | Simple answer                                                                                          |
|-------------------------------------|--------------------------------------------------------------------------------------------------------|
| Why use Spring Boot?                | It organizes the backend and makes routing, security, database access, and templates easier to manage. |
| How does it reduce plastic bottles? | People are more likely to refill if they can see which stations are available and trusted.             |
| What changed for guests?            | Guests now see a simple water guide instead of detailed maintenance data.                              |

## 15. Troubleshooting During Demo

- If login fails, check the username and password. Default admin is admin / admin123.
- If the app does not start, check that MySQL is running and the database password in application.properties is correct.
- If port 8080 is already used, run the app on another port, for example with --server.port=8081.
- If pages look old after a change, refresh the browser or restart the Spring Boot app.
- If a user cannot access a page, check their role. The app intentionally blocks pages based on role.

## 16. File Map for Quick Review

| Area                 | Files to mention                                                                                       |
|----------------------|--------------------------------------------------------------------------------------------------------|
| App start            | HydroTrackApplication.java                                                                             |
| Security             | SecurityConfig.java, UserService.java                                                                  |
| Authentication pages | AuthController.java, login.html, register.html, forgot-password.html                                   |
| Guest UI             | control-center.html, dashboard.html, fountains/list.html, fountains/detail.html                        |
| Admin users          | UserController.java, admin/users.html                                                                  |
| Data models          | User.java, Role.java, Fountain.java, WaterQualityRecord.java, FilterSchedule.java, MaintenanceLog.java |
| Database             | schema.sql, data.sql                                                                                   |
| Design               | style.css, app.js, refill-station.jpg                                                                  |

## 17. Final Summary

HydroTrack is more than a CRUD app. It connects user-friendly station discovery with staff-facing operational records. Guests get a simple guide to clean water. Staff get the tools to maintain trust in those stations. The database keeps a history, the backend protects actions by role, and the frontend communicates the clean-water mission clearly.